

2023



Red Barrels Launches The Outlast Trials with a Single Backend Engineer Using AWS

Learn how Red Barrels launched its first multiplayer game using Amazon GameLift.

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engineer built the game backend on AWS

10x

developer time savings on infrastructure management

50,000

concurrent players reached at early-ac



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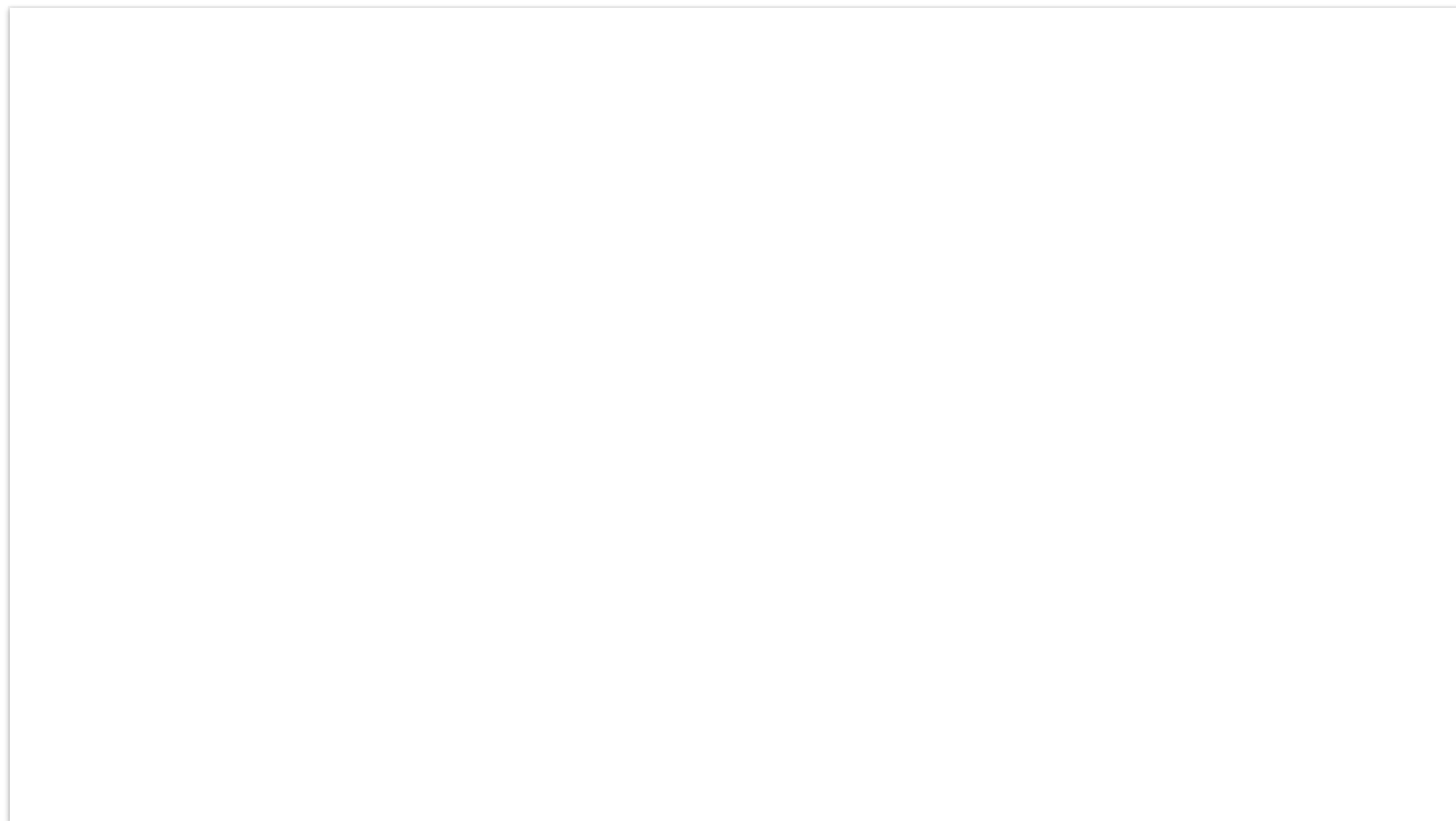
Overview

To create its first multiplayer game, Indie game studio [Red Barrels](#) set out to develop its recent... With no prior multiplayer experience and a small team, Red Barrels challenged itself to take a big step forward. The



studio hired a single backend developer to create the infrastructure for The Outlast Trials. “When we started developing this game, we had fewer than 15 people in the studio,” says Sébastien Sauvé, software architect at Red Barrels. “Everyone had a critical role to play in making the game what it is today.”

Red Barrels knew that this game needed a simple and reliable backend that would scale to meet player demand, so it developed the game on Amazon Web Services (AWS). After 5 years of hard work, the studio is proud of the game it has developed, which is set to launch in late 2023. “We feel we’re at the same level as any of the bigger studios,” says Sauvé. “And I built the backend with a team of one on AWS.”



Opportunity | Using Amazon GameLift to Scale Game Servers Reliably

Red Barrels is a fully independent studio based in Quebec, and it is known for its successful horror franchise, Outlast. All Outlast games had offered a rich single-player experience, but the studio aimed to shake things up with multiplayer play for The Outlast Trials. Yet, if the title was to succeed, the game needed to be highly reliable, simple to maintain, and cost efficient. To meet those requirements while preserving developer time for the game itself, Red Barrels decided to build its backend infrastructure entirely on AWS. “For us, using a lot of managed services made sense,” says Sauvé. “If you use AWS services properly, you reduce the time that developers spend managing infrastructure by 10 times.”

The Canadian game studio spent 5 years building The Outlast Trials. During that time, Sauvé spent only 30 percent of his time on backend work, devoting the rest to improving the actual game play. The first year of backend work was spent laying a robust foundation that could accommodate a wide range of game features. Red Barrels selected [Amazon GameLift](#), which deploys and manages dedicated game servers on AWS, to verify that its game scaled with fluctuating player demand. “We didn’t find anything that could solve as many challenges as Amazon GameLift,” says Sauvé. “It was one of the main reasons we went with AWS.”



Solution | Using Managed Services to Reduce Cost and Time

Initially, Red Barrels evaluated developing solutions on its own, but it wanted to save developer time. The company opted to use [Amazon DynamoDB](#), a fully managed key-value and document database, for most of its storage needs. To build a number of its game environments, Red Barrels implemented [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service for running Kubernetes. “The whole stack that we use is very simple,” says Sauv  . “Every time I add something, I have to think about maintaining it in the future.”

After the foundation was in place, it was time to start adding features. The Outlast Trials builds on the story arc of the other Outlast titles, and Red Barrels wanted the narrative to be just as compelling as those of the single-player games. But because the studio switched to a new game engine, Unreal Engine 4, everything about the technical aspects of the game was made from scratch. This challenged the team to develop creative features. “It’s never fun saying no to game designers who are trying to make the game fun,” says Sauv  . “But building new features was always possible and simple to do using what AWS offered.”

After a successful beta in 2022, Red Barrels extensively used [AWS Infrastructure Event Management](#) (AWS IEM), a service offering architecture and scaling guidance, to enhance the cost efficiency and performance of its game. Equipped with best practices, Red Barrels implemented a highly effective architecture that makes use of AWS tools. When the early-access release kicked off in the spring of 2023, a member of the AWS IEM team was present onsite, and several others were available on Discord, just in case an issue came up. “It’s great to have someone to contact directly,” says Sauv  . “Using AWS IEM, you have someone you can contact anytime, even at night.”

To the studio’s credit, there were no issues during the launch. The title had already sold one million copies after 2 months, with 50,000 concurrent players during the early-access weekend. Red Barrels is also hoping to set new records for Twitch viewership. “We’ve had no negative feedback in terms of game stability,” says Sauv  . “People are really happy about performance, connectivity, and gameplay.”

Outcome | Launching More Multiplayer Games on AWS

Red Barrels already has its eye on optimization. The studio built a custom analytics service to gameplay, and it wants to build on that solution. Much will depend on the full launch. “We’re success of our game so far, and we want to see how far this thing can go,” says Sauv  .



With its first multiplayer game ready to launch in late 2023, Red Barrels is excited to enter a new era of game development. “We built everything on AWS with the assumption that we’d have more multiplayer titles coming,” says Sauv  . “We hope our studio will grow and build lots of multiplayer games in the future.”

About Red Barrels

Indie game studio Red Barrels is an independently owned studio that was founded in 2011 in Quebec by three industry veterans. It is well known for its Outlast horror franchise.

AWS Services Used

Amazon GameLift

Amazon GameLift deploys and manages dedicated game servers hosted in the cloud, on-premises, or through hybrid deployments.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a key-value and document database that delivers single-digit millisecond performance at any scale.

[Learn more »](#)

AWS Infrastructure Event Management

AWS Infrastructure Event Management (IEM) offers architecture and scaling guidance and operational support during the preparation and execution of planned events, such as shopping holidays, product launches, and migrations.

[Learn more »](#)



Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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